

Nithya Balasundar

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SUMMARY

Senior Backend Engineer with 6+ years designing and scaling cloud-native distributed systems using Java, Spring Boot, Microservices, REST APIs, AWS, and Kubernetes. Proven record building high-throughput platforms (2M+ events/day, 1B+ records/month) with sub-second latency, and shipping production AI/LLM systems using LangGraph, RAG, embeddings, vector search, and multi-agent architectures. Skilled in system design, performance optimization, observability, CI/CD, and Agile delivery.

EXPERIENCE

Tata Consultancy Services (Client: Apple Inc.)

Software Engineer (Big Data), Backend & AI Platforms

Elk Grove, CA

Oct 2024 – Present

- Designed Java Spring Boot microservices processing 2M+ events/day for manufacturing operations, and built secure RESTful APIs for production tracking and inventory management with sub-second response times under high concurrent load.
- Architected scalable microservices using OOP principles and design patterns; developed high-throughput transactional services on Cassandra and PostgreSQL with event-driven architecture for improved scalability and fault tolerance.
- Containerized and deployed microservices on AWS EKS (Kubernetes) via automated CI/CD pipelines; optimized SQL queries and caching to reduce latency and cost, and integrated Apache Spark for large-scale batch analytics.
- Designed and built a multi-agent backend service (FastAPI + LangGraph) that translates natural-language analytics questions into charts, tables, and auto-generated Python widget code for Apple's IPA Studio platform.
- Architected a two-tier agent system, a LangGraph classification/planning agent plus a parallelized code-generation pipeline (plan, fan-out, validate/reconcile), that self-corrects Python widget code using Claude.
- Implemented RAG pipelines with embeddings and vector search for enterprise knowledge retrieval; integrated Claude and Gemini via an internal LLM gateway with mTLS/token-based auth and per-user usage-cost tracking.
- Built real-time SSE streaming with a dual-layer Redis and in-memory session cache for stateful, load-balanced multi-pod deployments; instrumented the pipeline with Langfuse tracing for LLM observability.
- Enhanced observability through centralized logging and monitoring (Kibana), reducing incident detection time by 40%; authored unit and integration tests (JUnit, Mockito) to improve deployment reliability.

Tata Consultancy Services (Client: Nielsen LLC)

Software Engineer – Java / API Developer

Florida

Sep 2021 – Sep 2024

- Developed Spring Boot microservices for enterprise media analytics platforms processing 1B+ records/month; built multi-threaded ingestion pipelines handling millions of events/day from streaming and batch sources.
- Improved reporting and dashboard performance 4x through SQL optimization, indexing, and caching strategies.
- Integrated Apache Spark via Apache Livy for near real-time analytics with end-to-end latency under five minutes; developed backend services supporting AI-assisted, natural-language querying.
- Resolved production concurrency and data-consistency issues in distributed systems; orchestrated ETL workflows with Apache Airflow and CI/CD automation with Jenkins across AWS environments.

TECHNICAL SKILLS

Languages: Java, Scala, Python, SQL

Backend & API Development: Spring Boot, Spring MVC, REST APIs, Microservices, JDBC, Hibernate, Multithreading & Concurrency, System Design, Design Patterns, OOP

AI / LLM Engineering: LLMs, LangGraph, Claude, Gemini, Retrieval-Augmented Generation (RAG), Prompt Engineering, Embeddings, Vector Search, Semantic Search, AI Agents, Langfuse

Cloud & DevOps: AWS (EKS, EMR, S3, Lambda, Kinesis, Glue), Kubernetes, Docker, Jenkins, Git, CI/CD, mTLS, Token-Based Auth

Data & Databases: Apache Spark, Apache Iceberg, Airflow, Event-Driven Architecture, ETL, Batch & Streaming Processing, Cassandra, MySQL, PostgreSQL, Redis

Testing & Practices: JUnit, Mockito, Kibana, Unit & Integration Testing, Agile/Scrum, Performance Optimization, Cross-Functional Collaboration

EDUCATION

University of Texas at Arlington

M.S., Computer Science & Engineering — GPA: 3.83/4.0

Arlington, TX

2021

Certifications: AWS Certified Cloud Practitioner | Google Cloud Data Engineering